

ROBOTIC SURGERY / MEDTECH FIELD OPERATIONS

Bottava Predictive Maintenance

Predict which surgical robots need service — before a failure interrupts a case. One governed, AI-powered workflow from telemetry to technician action.

8

Robots monitored across the surgical fleet

24

Components scored for maintenance risk

8 signals

Operational inputs the ML model weighs per component

THE SOLUTION

From reactive repairs to predictive service

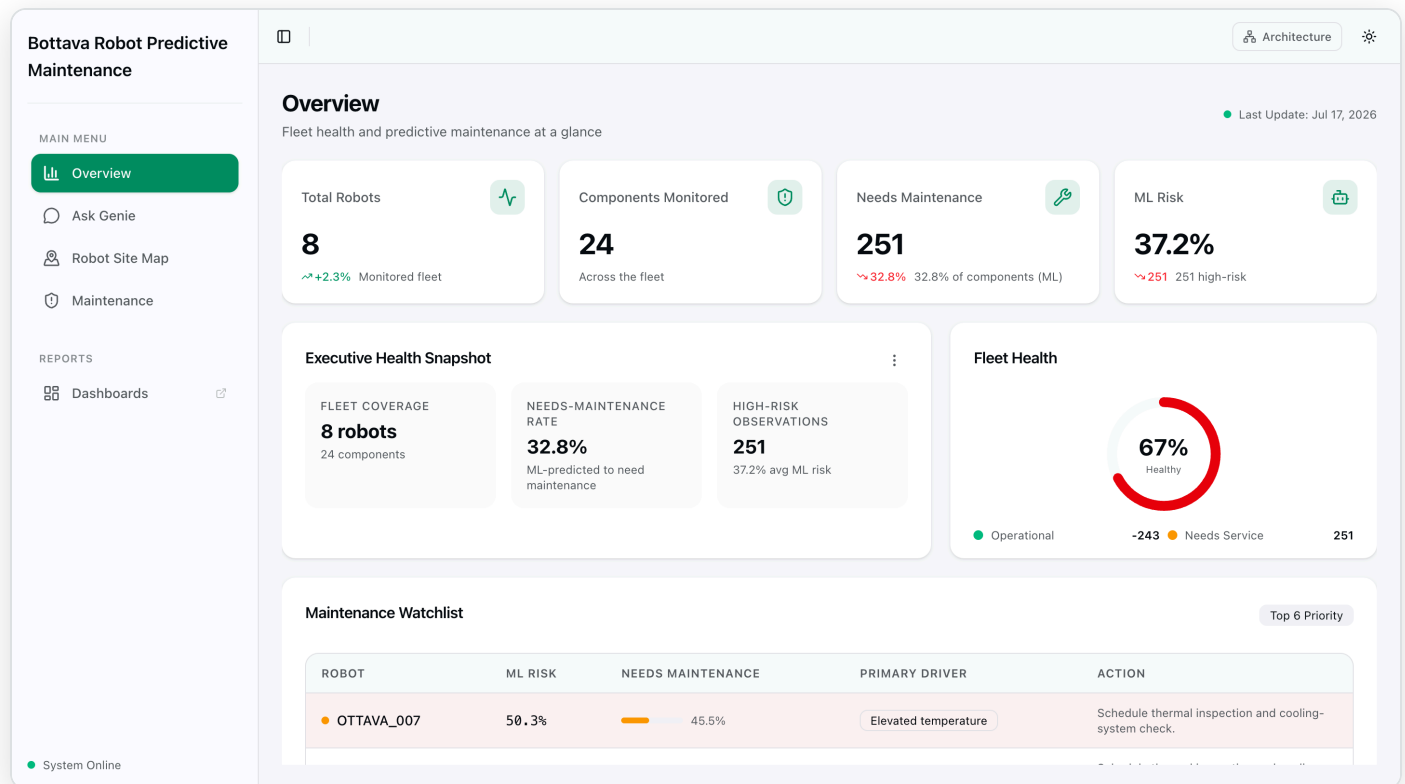
Bottava Predictive Maintenance unifies fleet telemetry, surgical-case history, and site data into one Unity Catalog–governed lakehouse, scores every robot component with a machine-learning model, and surfaces the results in an operator app — so field teams intervene on the right robot at the right time.

THE CHALLENGE

- Reactive service means unplanned downtime that disrupts surgical schedules.
- Fleet telemetry, case history, and site data live in disconnected silos.
- Teams lack an early-warning system to prioritize which robots to service first.

KEY CAPABILITIES

- Component-level ML maintenance risk score for every robot
- Prioritized watchlist that ranks the fleet by ML risk
- Geographic site map for location-aware triage
- Component heatmap isolating stressed parts
- Ask-in-plain-English analytics + AI recommended actions



Fleet overview: ML risk, needs-maintenance rate, executive health snapshot, and prioritized watchlist.

AI-POWERED INSIGHT

A machine learning model weighs eight operational signals — errors, thermal stress, vibration, utilization, and service history — to score every component's maintenance risk. An AI copilot then explains each

HOW IT WORKS

An end-to-end predictive workflow

1 Ingest & unify

Lakeflow pipelines land robot telemetry, surgical-case records, and site locations into bronze → silver → gold tables, all governed by Unity Catalog.

2 Score with ML

A model registered in Unity Catalog (MLflow Model Registry, Champion alias) runs batch inference over eight operational features to produce a maintenance-risk probability for every component-day.

3 Serve to operators

Gold results sync to Lakebase (Postgres) and power a Databricks App — watchlist, site map, and component heatmap — for fast, location-aware triage.

4 Explain & act

Mosaic AI Foundation Model API generates plain-language rationales and recommended actions; a Genie space answers ad-hoc fleet questions in natural language.

BUSINESS VALUE**Predict, don't react**

Shift from failure-driven repairs to proactive, scheduled intervention across the fleet.

One governed workflow

Ingestion, ML, dashboards, and an operator app run on a single Unity Catalog-governed platform.

Explainable AI

Every risk score comes with a plain-language rationale and recommended maintenance action.

DATABRICKS PLATFORM

Lakeflow Pipelines

Unity Catalog

MLflow

Model Registry

Mosaic AI

Foundation Model API

Genie

Lakebase

SQL Warehouse

Databricks Apps

ARCHITECTURE REFERENCE

Data sources to applications — one lakehouse

Unity Catalog End-to-end governance, lineage & access control across every table, model, and app**DATA SOURCES**

Robot telemetry (errors, temperature, vibration)

Surgical case history & outcomes

Robot asset & service records

Site locations & regions

**DATABRICKS PLATFORM****LAKEHOUSE**

Lakeflow Pipelines (bronze → silver → gold)

Delta Lake · SQL Warehouse

MACHINE LEARNING

MLflow · Model Registry (Champion)

Batch inference — component risk score

GENERATIVE AI

Mosaic AI Foundation Model API

Genie natural-language analytics

SERVING

Lakebase (Postgres) synced tables

**APPLICATIONS**

Fleet Overview & health snapshot

Maintenance watchlist (ML risk)

Robot site map

Component heatmap

Ask Genie chat

AI maintenance analysis

DEPLOYMENT

Databricks Asset Bundle — jobs, dashboards & app deployed as one governed bundle.

AUTHENTICATION

OAuth via Databricks Apps; service principal scoped with Unity Catalog & Lakebase grants.

TIME TO DEPLOY

Reusable accelerator — clone the bundle, point at your catalog, and deploy.